

Data Collection and Preprocessing Phase

Date	12 August 2026
Project Title	AI Career Pulse: Decoding the Job Market

Data Quality Report

The AI job-market dataset was reviewed for quality during the preliminary phase prior to Tableau analysis. The data quality assessment below was performed on 1500 records.

Data Source	Data Quality Issue	Severity	Resolution Plan
AI Job-Market Dataset	Missing values: 25 fields in the AI Job Market excel sheet have no missing cells.	Low	No missing values require treatment or imputation for this data set.
AI Job-Market Dataset	Duplicates: 0 rows with duplicate values and 0 duplicate values for the field job id were identified.	Low	There are no duplicate records to be deleted for this data set.
AI Job-Market Dataset	Experience consistency: 1099 out of 1500 records have a mismatch between the experience_level and years_of_experience. For example, AIJOB0002 is tagged as 'Senior (6-9 yrs)' but has 2 years of experience.	High	The years of experience should be used to calculate experience level. The experience level flag should be standardized to align with the ranges of years of experience, for instance: 0-2 Entry, 3-5 Mid, 6-9 Senior, and 10+ Lead. The years of experience may need to be used to standardize other fields, e.g., experience flag.
AI Job-Market Dataset	Salary range consistency: 288 records (19.2%) have annual salary usd higher than salary max usd. For example, AIJOB0003 has an annual_salary usd of \$360,000 while salary max usd = \$300,000.	Moderate	It is recommended to check the source definition of annual salary usd versus salary min usd/salary max usd to identify whether salary max usd is not a typo and address the inconsistency using the source control logic if the fields mean the same salary range.
AI Job-Market Dataset	Salary-tier consistency: 36 records have values of annual salary usd that do not align with the salary-tier ranges specified in the description. For example, there are entries with an annual salaryusd of \$100,000 are labelled 'Entry (<\$100k)'.	Moderate	It is recommended to standardize the salary-tier boundaries and recalculate the salary tier from the annual salary using one documented rule before proceeding with dashboard preparation in Tableau.

AI Job-Market Dataset	Range validation: demand score is 68-98, benefits score_10 is 6.0-9.8, posting month is 1-12, and posting year is 2025-2026. There are no values that exceeded the defined ranges for these fields.	Low	No action is required to address range validation since no values exceeded the defined ranges for these fields.
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Data Cleaning Process

Cleaning Area	Cleaning Process
Missing values	Review all 25 fields for missing values that would require further treatment or imputation. At this stage, no missing data was identified.
Duplicates	Check all fields for duplicates. At this stage, no duplicated records were identified.
Experience fields	Validate the consistency of experience level versus years_of_experience and standardize experience_level if needed.
Salary fields	Ensure that the annual_salary_usd fields are consistent with salary_min_usd/salary_max_usd and clarify the source rule if needed.
Salary tier	Recalculate/standardize the salary tier using the annual salary as per the field description.
Final validation	Review fields again after revisions and confirm that no data quality issues were identified. The dataset is ready for Tableau.